Lecture 04, Math 441

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Exact Equations (cont.)

Ex: Rewrite $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-3x^2-2xy}{x^2+2y} \rightarrow (x^2+2y)\mathrm{d}y = (-3x^2-2xy)\mathrm{d}x$

$\rightarrow \underbrace{(3x^2+2xy)}_{M}\mathrm{d}x + \underbrace{(x^2+2y)}_{N}\mathrm{d}y = 0$ (*)

Define $M(x,y) = (3x^2+2xy)$ and $N(x,y) = (x^2+2y)$

Check that $M_y = N_x$:

$$M_y = 2x; \qquad N_x = 2x$$

So $\boxed{M_y = N_x}$ and the equation is exact, in other words rewrite (*)

as $\mathrm{d}\Psi = 0$ such that $\Psi_x = M$ and $\Psi_y = N$.

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To find Ψ we start by integrating $\Psi_x = M = 3x^2 + 2xy$ in x:

$$\Psi(x,y) = \int M(x,y)\mathrm{d}x + h(y) = x^3 + x^2y + h(y)$$

Then we use $\Psi_y = N$: $\Psi_y = \cancel{x^2} + h'(y) = \cancel{x^2} + 2y$

So $h'(y) = 2y$ and $h(y) = y^2$.

Therefore $\Psi(x,y) = x^3 + x^2y + y^2$

$\mathrm{d}\Psi = 0 \quad \rightarrow \quad \Psi(x,y) = A$

and the solution to the ODE is $\boxed{x^3 + x^2y + y^2 = A}$

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Ex: $\underbrace{(2\ln(y) + 3xe^y)}_{M}dx + \underbrace{(\frac{x}{y} + x^2e^y)}_{N}dy = 0$

Call $M(x,y) = (2\ln(y) + 3xe^y)$ and $N(x,y) = (\frac{x}{y} + x^2e^y)$ and

check that $M_y = N_x$:

$$M_y = \frac{2}{y} + 3xe^y; \qquad N_x = \frac{1}{y} + 2xe^y$$

So $M_y \neq N_x$ and the equation is not exact.

What if we tried to find Ψ anyway?

Set $\Psi_x = M$ and $\Psi_y = N$

$\Psi_x = 2\ln(y) + 3xe^y$

$\Psi = \int M\mathrm{d}x + f(y) = 2x\ln(y) + \frac{3}{2}x^2e^y + f(y)$

Then $\Psi_y = \frac{2x}{y} + \frac{3}{2}x^2e^y + f'(y) = \frac{x}{y} + x^2e^y$

So that $f'(y) = -\frac{x}{y} - \frac{1}{2}x^2e^y$ which doesn't work since f is not

supposed to depend on x.

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Let's try using an integrating factor $\mu(x, y)$. Multiply by $\mu(x, y)$

and get: $\underbrace{\mu(2\ln(y) + 3xe^y)}_{M(x,y)}dx + \underbrace{\mu\left(\frac{x}{y} + x^2e^y\right)}_{N(x,y)}dy = 0$

Impose the condition for the equation to be exact:

$M_y = N_x$

$M_y = \mu_y(2\ln(y) + 3xe^y) + \mu(\frac{2}{y} + 3xe^y)$

$N_x = \mu_x(\frac{x}{y} + x^2e^y) + \mu(\frac{1}{y} + 2xe^y)$

$\mu_y(2\ln(y) + 3xe^y) + \mu(\frac{2}{y} + 3xe^y) = \mu_x(\frac{x}{y} + x^2e^y) + \mu(\frac{1}{y} + 2xe^y)$

This reduces to:

$\mu_y(2\ln(y) + 3xe^y) - \mu_x\left(\frac{x}{y} + x^2e^y\right) + \mu\left(\frac{1}{y} + xe^y\right) = 0$

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This can be made easier to solve with the assumption $\mu = \mu(x)$,

i.e. $\mu_y = 0$.

$$\cancel{\mu_y(2\ln(y) + 3xe^y)} + \mu(\tfrac{2}{y} + 3xe^y) = \mu_x(\tfrac{x}{y} + x^2e^y) + \mu(\tfrac{1}{y} + 2xe^y)$$

We are left with

$$\mu_x\left(\tfrac{x}{y} + x^2e^y\right) = \mu\left(\tfrac{1}{y} + xe^y\right)$$

$$\mu_x\cancel{(\tfrac{1}{y} + xe^y)}x = \mu\cancel{(\tfrac{1}{y} + xe^y)}$$

or simply $\mu_x x = \mu \quad \rightarrow \quad \int \frac{d\mu}{\mu} = \int \frac{dx}{x} \quad \rightarrow \quad \mu = \not{c}x$

and has the solution $\boxed{\mu(x) = x}$

This integrating function should make the original equation exact.

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Let's try. Multiply the equation by $\mu(x) = x$ and get

$$\underbrace{(2x\ln(y) + 3x^2 e^y)}_{M}\mathrm{d}x + \underbrace{(\frac{x^2}{y} + x^3 e^y)}_{N}\mathrm{d}y = 0.$$

Now check the condition for exactness: $M_y = N_x$

$$M_y = \left(2x\ln(y) + 3x^2 e^y\right)_y = \frac{2x}{y} + 3x^2 e^y$$

$$N_x = \left(\frac{x^2}{y} + x^3 e^y\right)_x = \frac{2x}{y} + 3x^2 e^y$$

So the equation is indeed exact now.

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So let's find the solution to the equation now.

$$\Psi_x = M = 2x \ln(y) + 3x^2 e^y$$

$$\Psi = x^2 \ln(y) + x^3 e^y + f(y)$$

$$\Psi_y = \frac{x^2}{y} + x^3 e^y + f'(y) = \frac{x^2}{y} + x^3 e^y$$

This means $f'(y) = 0$ so $f(y) = c$ is a constant, which we can just choose to be 0. So now

$$\Psi(x, y) = x^2 \ln(y) + x^3 e^y$$

$$\Psi(x, y) = A$$

and the solution is $\boxed{x^2 \ln(y) + x^3 e^y = A}$

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Substitution Methods

Consider the equation $x\frac{dy}{dx} = \frac{y^2}{x} + y$

$$x\,dy = \left(\frac{y^2}{x} + y\right)dx \qquad \underbrace{\left(-\frac{y^2}{x} - y\right)dx}_{M} + \underbrace{x}_{N}dy = 0$$

$M_y = \frac{-2y}{x} - 1 \qquad N_x = 1$

$\frac{dy}{dx} = \frac{y^2}{x^2} + \frac{y}{x}$

Try the substitution $v(x) = \frac{y(x)}{x}$. So $y = xv$ and

$y' = \frac{d}{dx}(xv(x)) = v + xv'$ (derivative of a product)

Substituting into the original equation we get

$v + xv' = v^2 + v$

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This is a separable equation: $x\frac{\mathrm{d}v}{\mathrm{d}x} = v^2 + v - v$

$$\rightarrow \int \frac{\mathrm{d}v}{v^2+v-v} = \int \frac{1}{x}\mathrm{d}x$$

and solving it we get

$$\int \frac{\mathrm{d}v}{v^2} = \int \frac{\mathrm{d}x}{x} \rightarrow -\frac{1}{v} = \ln(x) + A \rightarrow v(x) = \frac{1}{A - \ln(x)}$$

Don't forget to go back to the original unknown $y(x)$:

$$\boxed{y(x) = \frac{x}{A - \ln(x)}}$$

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In general this substitution works whenever we can rewrite the ODE as

$$\frac{\mathrm{d}y}{\mathrm{d}x} = F\left(\frac{y}{x}\right)$$

Use $v(x) = \frac{y(x)}{x}$, $y(x) = xv(x)$, and $y' = xv' + v$

and obtain a separable equation for $v(x)$.

Can even use this for something like: $\frac{\mathrm{d}y}{\mathrm{d}x} = \sin\frac{y}{x} + \mathrm{e}^{\frac{y}{x}} + \frac{y^3}{x^3} + \frac{x}{y} + 2$

Another ex: $x\,y\,y' = y^2 + x\sqrt{4x^2 + y^2}$

Rewrite as $y' = \frac{y}{x} + \sqrt{4\frac{x^2}{y^2} + 1} = F\left(\frac{y}{x}\right)$

So we use the substitution $v = \frac{y}{x} \to y = xv \to y' = v + xv'$:

$$\cancel{v} + xv' = \cancel{v} + \sqrt{\frac{4}{v^2} + 1}$$

$$\to x\,v' = \sqrt{\frac{4}{v^2} + 1}$$

which is separable.

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Separate and integrate:

$$\frac{\mathrm{d}v}{\sqrt{\frac{4}{v^2}+1}}\mathrm{d}v = \frac{\mathrm{d}x}{x} \;\rightarrow\; \int \frac{v}{\sqrt{4+v^2}}\mathrm{d}v = \int \frac{\mathrm{d}x}{x}$$

$$\rightarrow \sqrt{4+v^2} = \ln(x) + A \qquad \rightarrow \qquad 4 + v^2 = (\ln(x) + A)^2$$

and going back to $y(x)$:

$$\boxed{y^2 = x^2\left(\ln(x) + A\right)^2 - 4x^2}$$

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